# Resolution Geometry

## The Stress Tensor: Tensor Descent, Two Routes, and the Field Equation as Consilience

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**Abstract**

We make the stress–energy tensor an RG target. A tensor-valued task is LOCKED exactly when the observation kernel is contained in its kernel, and it is a CORRESPONDENCE set otherwise. There are two independent routes to the tensor:

- a *geometric* route, through the Einstein tensor of a metric resolved to second order;
- a *source* route, through direct field stresses or the directional moments of matter, $T^{ab} = \int p^a p^b f \, d\Pi$.

Their agreement is the field equation, read as a consilience test, and their disagreement locates the failing assumption.

Exact symbolic computation establishes the minimal package:

1. The Schwarzschild exterior has curvature (Kretschmann $48M^2/r^6$) but $T = 0$, so no curvature or Fisher scalar is a local energy density.
2. The Reissner–Nordström geometry and the Maxwell stress of the Coulomb field give the same tensor, and a source declared with the wrong charge is INCONSISTENT.
3. On FLRW, radiation and dust source routes agree with the geometric route at $\alpha = \frac{1}{2}$ and $\frac{2}{3}$, and the swapped assignment is inconsistent.
4. The background $(\epsilon, p)$ does not identify its source: a massless scalar and a stiff fluid coincide.
5. The radiation stress given only the energy density is exactly the set $\{S \succeq 0,\ \operatorname{tr} S = \epsilon\}$.
6. The stress depends only on the $\ell \leq 2$ angular harmonics, so $\ell \leq 2$ is the minimal sufficient panel.
7. Metric agreement does not control stress: perturbations with sup $\to 0$ carry stress $\to \infty$.
8. Under isotropic rescattering, the stress anisotropy is forgotten exactly as $(1-s)^n$, uniformly over every declared direction law.

We do *not* show that general relativity does not need spacetime. The geometric route uses a metric, which RG can derive from observations under its own hypotheses, and an imported field equation, which Lovelock's theorem singles out under stated premises. What is shown is that stress and geometry are both certifiable RG observables, and that the Einstein equation is a testable relation between them. Eleven checks pass on five seeds, and three canaries are caught.

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# 1 The target and the two routes

**Definition 1.1** (Tensor-valued task)**.** Let $\mathcal{O}(\theta)$ be the joint observation law and $\mathcal{T}(\theta) = T^a{}_b$ the stress tensor, given in a declared frame or by its invariants (eigenvalues of $T^a{}_b$).

**Theorem 1.2** (Tensor descent; ladder L2 [1])**.** *The task is LOCKED iff* $\mathcal{O}(\theta_1) = \mathcal{O}(\theta_2) \Rightarrow \mathcal{T}(\theta_1) = \mathcal{T}(\theta_2)$*. Otherwise the RG output is the correspondence* $\mathfrak{T}(q) = \{\mathcal{T}(\theta) : \mathcal{O}(\theta) = q\}$*. Componentwise, exactly the components (or invariants) that factor through the quotient are identified. Frame-dependent components need a declared frame; the invariants do not.*

*Proof.* This is universal factorization (L2) applied to each component, and to the invariants under the frame group. □

**The geometric route.** $T^a{}_b = G^a{}_b[g]/8\pi$ in geometrized units, with $\Lambda$ by a declared convention. It needs a metric resolved in $C^2$ and the field equation.

**The source route.** $T^{ab}$ is computed from the source itself: the Maxwell stress of a measured field, or the kinetic moments $\int p^a p^b f \, d\Pi$ of a measured distribution. No metric derivative enters.

**Imported Theorem 1.3** (Lovelock [4])**.** *In four dimensions, the only symmetric, divergence-free 2-tensors built locally from the metric and its first two derivatives are* $aG_{ab} + bg_{ab}$*.*

The geometric route is therefore not a free choice: it is selected by locality, second order and conservation, just as Fisher is selected by Markov invariance. Its constants are fixed by anchors (Newtonian limit) and by the $\Lambda$ convention.

# 2 No-gos

*No-Go* 2.1 (Curvature is not stress; ST02). In the Schwarzschild exterior, every component of $G^a{}_b$ vanishes, while the Kretschmann scalar is $48M^2/r^6$. Tidal (Weyl) curvature is not stress; only the Ricci part is. Any map from a curvature, Fisher or coherence scalar to $T_{00}$ is falsified by the vacuum.

*No-Go* 2.2 (Energy does not determine stress; ST06). Isotropic radiation ($T^{ij} = \frac{\epsilon}{3}\delta^{ij}$) and counter-propagating beams ($T^{ij} = \epsilon\,\mathrm{diag}(1,0,0)$) share energy and zero flux. For radiation of energy $\epsilon$, the correspondence is exactly

$$\mathfrak{T}(\epsilon) = \{S \in \mathrm{Sym}_3 : \ S \succeq 0, \ \mathrm{tr}\, S = \epsilon\}.$$

Every direction law lands in this set, and every member is realized by six beams along its eigenvectors (Carathéodory; realization error below $10^{-12}$).

*No-Go* 2.3 (Metric agreement does not certify stress; ST08). Perturbations $\eta \sin(r/\eta^2)$ of $f(r)$ have $\sup \to 0$ ($10^{-3}$), while the stress they induce grows like $\eta^{-3}$ (above $10^6$). The geometric route needs a $C^2$ certificate at the declared scale.

**Proposition 2.4** (The background does not identify the source; ST05)**.** *At* $a \propto t^{1/3}$*, the massless scalar* $\phi = \log t/\sqrt{12\pi}$ *(with* $V = 0$ *and a field-equation residual of* 0*) and a stiff fluid give the same* $(\epsilon, p)$*. The source is a CORRESPONDENCE until another observable separates the candidates: perturbations, sound speed or couplings.*

## 3 Locks

**Theorem 3.1** (Minimal angular panel; ST07)**.** *For massless radiation with intensity $I(\mathbf{n})$,*

$$T^{00} = \int I \, d\Omega, \qquad T^{0i} = \int n^i I \, d\Omega, \qquad T^{ij} = \int n^i n^j I \, d\Omega.$$

*These depend only on the harmonics $\ell \le 2$, so every $\ell \ge 3$ harmonic is in the kernel of the stress map. The minimal sufficient angular panel is therefore the monopole, dipole and quadrupole.*

*Proof.* $n^i n^j$ spans only the $\ell = 0$ and $\ell = 2$ harmonics, and $n^i$ spans only $\ell = 1$; orthogonality of spherical harmonics does the rest. Numerically, adding $\ell = 3..6$ components changes the stress by less than $10^{-4}$ (quadrature level), while an $\ell = 2$ change moves it by more than $10^{-2}$. □

**Theorem 3.2** (Two routes, one tensor; ST03, ST04)**.**

1. Reissner–Nordström. *With $m = M - Q^2/2r$, the geometric route gives $\epsilon = Q^2/8\pi r^4$, $p_r = -\epsilon$ and $p_\perp = +\epsilon$. The Maxwell stress of $E = Q/r^2$, which involves no metric derivative, gives the same tensor exactly. A source declared with $1.1\,Q$ is INCONSISTENT.*

2. Flat FLRW, $a = t^\alpha$. *The geometric route gives $\epsilon = 3H^2/8\pi$, $p = -(2\dot{H} + 3H^2)/8\pi$ and $w = 2/(3\alpha) - 1$, with conservation holding identically. The kinetic radiation route ($w = \frac{1}{3}$) agrees at $\alpha = \frac{1}{2}$, and dust ($w = 0$) agrees at $\alpha = \frac{2}{3}$. The swapped assignment is INCONSISTENT.*

The Einstein tensor was computed independently (Christoffel symbols, then Riemann, Ricci and Einstein [5, 6]), and it reproduces $G^t{}_t = G^r{}_r = (rf' + f - 1)/r^2$ and $G^\theta{}_\theta = f''/2 + f'/r$ exactly (ST01).

**Theorem 3.3** (Forgetting certificate for stress; ST09; [3])**.** *Under rescattering into an isotropic direction with probability $s$ per step (energy conserved),*

$$\mathbb{E}[T_n] = (1-s)^n T_0 + \big(1 - (1-s)^n\big)\tfrac{\epsilon}{3} I$$

*for* every *initial direction law. This identity is checked symbolically. Hence, uniformly over all declarations,*

$$\|T_n - \tfrac{\epsilon}{3} I\| \le \tfrac{2}{3}\epsilon(1-s)^n,$$

*and the beam declaration attains the bound within Monte Carlo error. Because the stress of radiation is bounded by $\epsilon$, the bound needs no further moment class. Unbounded stress functionals would need declared moment control.*

Forgetting certifies the *robustness* of the stress: it becomes independent of the declared initial state. It creates no directional information that was never observed.

**Proposition 3.4** ($\Lambda$ bookkeeping; ST10)**.** *Writing $G + \Lambda g = 0$, or $G = 8\pi T_\Lambda$ with $\epsilon_\Lambda = \Lambda/8\pi = -p_\Lambda$, solves the same equations exactly. The convention is a correspondence of descriptions, not of physics, and the certificate records which one is used.*

## 4 The RG stress certificate

| target | status | conditional on |
|---|---|---|
| Reissner–Nordström exterior stress | LOCKED (cond.) | Einstein equation; static spherical ansatz; metric in $C^2$; Maxwell source; units |
| flat FLRW $(\epsilon, p)$ | LOCKED (cond.) | Einstein equation; FLRW ansatz; $a(t)$ in $C^2$; kinetic source model |
| Schwarzschild exterior matter stress | LOCKED (cond.), $T = 0$ | Einstein equation; $\Lambda$ convention |
| radiation stress from the $\ell \le 2$ moments | LOCKED | massless kinetic model |
| radiation stress from energy only | CORRESPONDENCE | $\{S \succeq 0,\ \mathrm{tr}\, S = \epsilon\}$ |
| FLRW source identity | CORRESPONDENCE | $(\epsilon, p)$ only: scalar or stiff fluid |

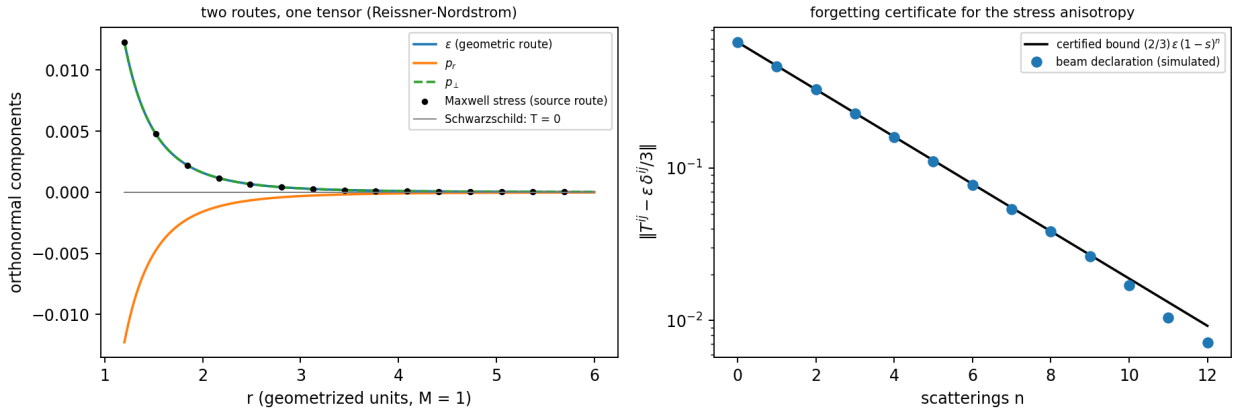



Figure 1: Left: the geometric route (curves) and the Maxwell source route (dots) give one tensor for Reissner–Nordström; Schwarzschild has $T = 0$. Right: the certified forgetting bound for the stress anisotropy and the beam declaration attaining it (this figure uses fewer photons than the battery).

This is the "RG stress tensor": a typed certificate, not a single selected tensor. It is LOCKED where two independent routes agree or the relevant moments are resolved, and it is an explicit set where they are not.

## 5 What this does and does not show about spacetime

- *Not shown: that GR does not need spacetime.* The geometric route uses a metric and its second derivatives. The field equation is imported, and selected by Lovelock's theorem under locality, second order and conservation. RG does not derive it.
- *Shown: spacetime need not be a primitive input.* The metric that the geometric route uses can be an RG *output* under named hypotheses:
  - the conformal class from a resolved cone (L64–L68);
  - the scale from counted channels and anchors (L94–L109);
  - spatial geometry on stationary slices (L104–L105).
- *Shown: the Einstein equation is a testable relation between two certified observables.* It is a consilience between a geometry route and a source route [2]. RG certifies it where both are resolved, and it locates the failing assumption where they disagree.
- *Shown: a stress tensor can be an RG certificate.* It is LOCKED, a CORRESPONDENCE set, or forgetting-certified, with its residue explicit.

## 6 Programme

1. *Real data, vacuum.* Solar-system ephemerides test $T = 0$ (Schwarzschild-type exteriors) against an independent source route (planetary masses from satellite dynamics).
2. *Real data, cosmology.* $H(z)$ gives $(\epsilon, p)$ through the geometric route, with a $C^2$ certificate for $\dot{H}$. Radiation and baryon densities from independent probes give the source route. Agreement or a typed residue follows.
3. *Stress from directional data.* Angular-resolved intensity up to $\ell = 2$ locks radiation stress. For massive matter, momentum-magnitude moments must also be declared or measured.

4. *Derivative certificates.* A resolution-indexed $C^2$ gate (STRUCTURAL WALL[$\ell$]) is needed for every geometric-route claim.

5. *The physical-curvature ladder.* The vacuum no-go separates Weyl curvature, which is visible tidally but carries no stress, from Ricci curvature, which is stress. Both are protocol-invariant physical curvature, distinct from RG's observable and latent connection curvatures.

# A Reproducibility

`rg_stress_attack.py` (ST01–ST11) passes 11/11 on seeds 20261025, 1, 42, 777 and 2026. The curvature checks are exact (SymPy). Its canaries are:

- treating Schwarzschild curvature as stress;
- a wrong source charge;
- an overclaimed forgetting rate.

All are caught. One correction was made before registration: ST09 first used a fixed ratio tolerance that ignored Monte Carlo noise once the bound fell to about $10^{-2}$. It now checks the exact operator identity symbolically and uses a noise-derived tolerance. The geometric formulas are exact computations under declared ansätze; they are not a historical-data analysis.

# References


[1] M. Rada, *Resolution Geometry* (print edition), 2026.

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[5] C. W. Misner, K. S. Thorne, J. A. Wheeler, *Gravitation*, Freeman, 1973.

[6] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.